WASP-49b

R-1020

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-49_221223` · created 2026-07-30T04:31:49+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459936.7923 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1401 +/- 0.0077
Transit depth (Rp/Rs)^2	1.96 +/- 0.22 [%]
Orbital Inclination (inc)	84.86 +/- 0.83
Airmass coefficient 1 (a1)	1.099 +/- 0.014
Airmass coefficient 2 (a2)	-0.055 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.24 %
Best Comparison Star	#3 - [532, 231]
Optimal Aperture	5.03
Optimal Annulus	6.11
Transit Duration (day)	0.097 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1401 ± 0.0077 vs 0.116 (deviation 2.32σ)
Duration fit vs published	0.097 d vs 0.09 d (deviation 0.47σ)
Mid-time O–C	-4.2 min
Depth SNR	13.5
Residual scatter	1.24 %

Light curve

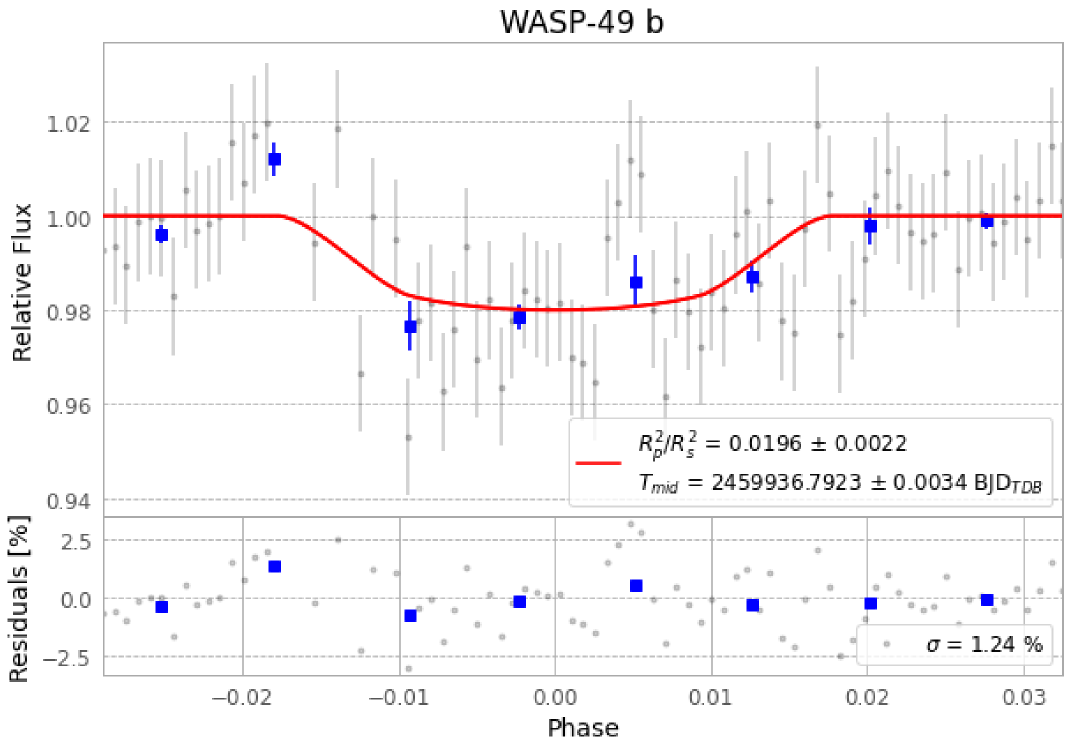

Figure 1 — FinalLightCurve_WASP-49 b_2022-12-22.png (SHA-256 648b9fbe9812c05d…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [530, 95], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 c54cc2bfc543b4f6…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

83 FITS frames (55 MB), `WASP-49221223045715.FITS` → `WASP-49221223090315.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-49-221223.log (2ff31ff02a2e…), FinalLightCurve_WASP-49 b_2022-12-22.png (648b9fbe9812…), FinalLightCurve_WASP-49 b_2022-12-22.csv (94a8f0c24623…)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-30 04:31Z.